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How many turkeys do we kill for Thanksgiving?

by Harish

Posted on November 18, 2015

The Thanksgiving season is upon us and, horrifically, upon turkeys. We will see their carcasses everywhere — sliced and served on the lunch plate, or dressed and stuffed on the dinner table. They will be talked about at the water-cooler and laughed about on late-night TV. There will even be turkey-flavored potato chips and turkey-infused vodka.

What must be the scale of an annual ritual concerning one species of birds that they will so invade our lives for a week or two? Exactly how many turkeys do we Americans kill for Thanksgiving?

The National Turkey Federation did estimate an answer to this question and came up with **46 million**, but provided no methodology or reasoning behind their estimate. This blog post fills the gap but arrives at a more conservative number of 37 million as likely closer to the correct answer.

From the monthly **USDA reports on poultry slaughter**, we know that we kill over 235 million turkeys each year — in 2014, we killed 236.6 million of them. But, it is not entirely straightforward to estimate the number of them we kill for our consumption during the Thanksgiving season because turkeys are killed year-round at approximately the same rate. It makes business sense to do so — killing at a steady rate, without sudden spikes during the holiday season, uses the machinery, labor, and other infrastructure more efficiently.

Since we eat more turkeys around Thanksgiving despite not killing them at a higher rate during the season, most of the turkeys we eat at Thanksgiving are ones who are killed months earlier and kept frozen. The USDA keeps track of the stocks of meat kept frozen in public and private warehouses before they are moved for retail to grocery stores, restaurants, and food service companies. Based on the **USDA reports on cold storage**, the following bar graph shows the weight of whole turkey carcasses as well as parts of turkeys, such as breasts and legs, kept frozen in warehouses in the US on the last day of each month, from January 2013 to September 2015 (the most recent month for which data is available).

Turkey carcasses and parts in cold storage
at the end of each month (in millions of pounds)
from January 2013 to September 2015

October 9, 2012

Electoral politics and the meat industry

August 23, 2012

Meat consumption patterns by race and gender

July 12, 2012

Look who is talking animal welfare!

May 7, 2012

Meat industry advertising

February 6, 2012

How many animals does a vegetarian save?

January 9, 2012

The paradoxical increase in the use of primates in research

October 24, 2011

Is Vegan Outreach right about how many animals suffer to death?

September 1, 2011

The pain of animals used in research

August 5, 2011

Do pathogens in animal products cause deadlier illnesses?

July 19, 2011

Where is the cow whose milk we drink?

July 12, 2011

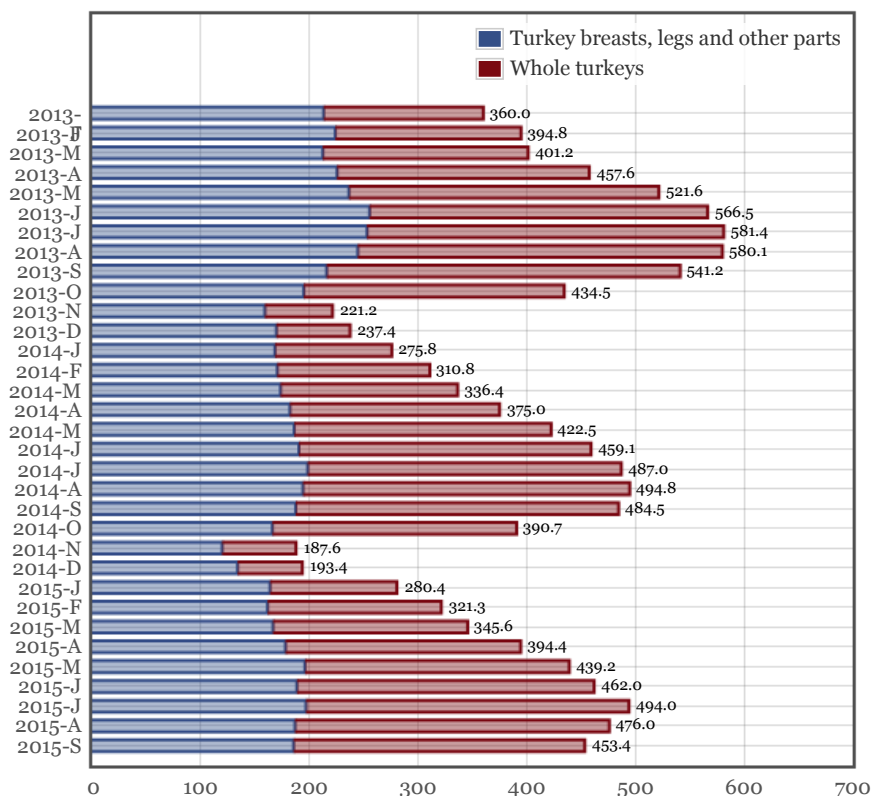
Vegan titles poised to overtake vegetarian?

July 11, 2011

Animal rights as contentious as the Israeli-Palestinian conflict?

“Not everything that can be counted counts, and not everything that counts can be counted.”

— William Bruce Cameron, 1963



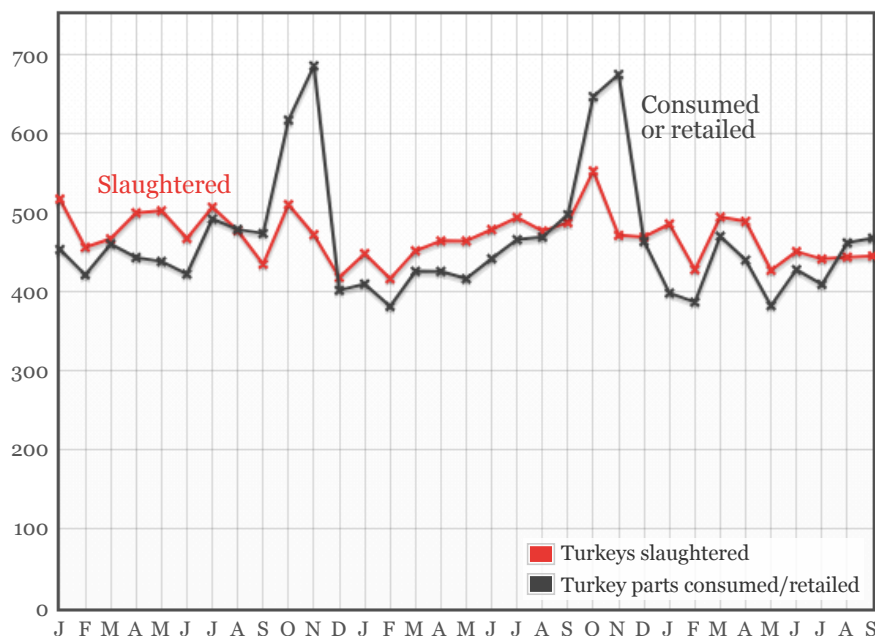
It is easy to notice the cyclic nature of the bar graph above — there are sharp drops in the amount stored in warehouses, especially of whole turkey carcasses, during October and November. These drops represent the movement of frozen turkeys from warehouses to grocery stores and other food establishments for sale or consumption. Timed for peak consumption in late November, this process begins in mid-October and lasts through mid-to-late November.

During any given month, the total weight of turkeys retailed (i.e., moved from cold storage for retail) or consumed can be estimated as the weight of turkeys killed during that month *plus* the weight of turkeys in cold storage on the last day of the previous month *minus* the weight of turkeys in cold storage on the last day of that current month. For example, the total weight of turkeys consumed/retailed during April is the weight of turkeys killed during April plus the weight of turkeys in cold storage on March 31 minus the weight of turkeys in cold storage on April 30.

The only little snag left is that the turkey slaughter data is reported in live weight while the cold storage data is reported in carcass weight or the actual weight of what is stored in the warehouse. The live weight of an animal is the weight of an animal when he/she is alive. The carcass weight or the dressed weight of a turkey is usually the weight of the carcass excluding the head, feet, and certain internal organs but including bones, skin, fat, liver, gizzard, and neck. The ratio of the carcass weight to live weight of a turkey varies between 77.74% to 80.03% depending on age, gender, and strain of the turkey. Since USDA reports on turkey slaughter and storage are not consistently separated by age, gender or strain, I will use the midpoint of this range, 78.885%, as the conversion factor from live weight to carcass weight.

The following graph displays the estimated carcass weight of the turkeys slaughtered during each month, between January 2013 and September 2015, and the estimated carcass weight of the turkeys consumed/retailed during each of those months.

Total weight of turkeys slaughtered and consumed/retailed
(in millions of pounds of estimated carcass weight)
in each month between January 2013 and September 2015



In the graph above, the additional weight of the turkeys consumed/retailed during October and November beyond what is typically consumed/retailed during the other ten months can be estimated as the carcass weight of the turkeys killed for Thanksgiving. During the last 24 months for which data is available (from October 2013 to September 2015), the average carcass weight of turkeys consumed/retailed per month, excluding October and November, was 431.96 million pounds. This would represent a baseline rate of consumption of turkeys if we did not have the Thanksgiving season. This consumption is usually in the form of turkey slices in sandwiches, turkey burgers, turkey sausages, and ground turkey, directly purchased by consumers or served at lunch delis and workplace cafeterias. For obvious economic reasons, these turkey products are made from the larger turkeys — ones with a carcass weight larger than the average of about 24 pounds.

The months of October and November, however, are different with a significantly higher rate at which turkeys, mostly smaller and whole turkeys, move out of warehouses and into supermarkets, food service stations, and restaurant menus. Since Thanksgiving appears in late November, it is a reasonable assumption that almost all of this inventory of turkeys — a total average of 1,310.9 million pounds during these two months — is intended for sale or consumption during the Thanksgiving season.

Now, the season itself is not just limited to the Thanksgiving week — as early as mid-November, food service cafeterias set up turkey carving stations on some days, restaurants add turkeys on their menus and workplace parties feature whole roast turkeys. I would conservatively estimate that turkey consumption during at least one-third of the month of November, or 10 days of November, is Thanksgiving-inspired consumption with little or no baseline consumption of the more processed turkey products like deli slices. So, the baseline consumption (not inspired by Thanksgiving) during October and November can be estimated at 431.96 million pounds during October plus two-thirds of 431.96 million pounds during November, a total of 719.94 million pounds. The additional weight of turkeys consumed/retailed for the Thanksgiving season per year, therefore, is $1,310.9 - 719.94 \approx 590.96$ million pounds.

Sources cited

1. National Agricultural Statistics Service, USDA. *Poultry Slaughter*. 2013-2015. ([link](#), accessed November 18, 2015)
2. National Agricultural Statistics Service, USDA. *Cold Storage*. 2013-2015. ([link](#), accessed November 18, 2015)
3. National Turkey Federation. *Turkey History and Trivia*. Undated. ([link](#), accessed November 18, 2015)
4. R. M. Hulet, Department of Poultry Science, Pennsylvania State University. *Comparative Meat Yield of Modern Commercial Turkeys*. *World Poultry magazine*, 2004. ([link](#), accessed November 18, 2015)

Based on a survey **reported** by the National Turkey Foundation, the average weight of turkeys purchased at Thanksgiving is 16 pounds — much smaller than the overall average carcass weight (24 pounds) of turkeys raised in the United States. The 590.96 million pounds of turkeys consumed/retailed for the Thanksgiving season, therefore, represents $590.96 / 16 \approx 36.9$ million turkeys.

We kill about 37 million turkeys for Thanksgiving.

It is a human weakness that we can be overwhelmed by the scale of the massacre these numbers represent, but lose sight of the suffering endured by each hen or tom who lies dead on our Thanksgiving table. From the time they were **babies in hatcheries** to the time **they were killed**, the turkeys we eat would have endured a horrific litany of abuses.

Our expression of gratitude for the joys in our lives need not come coupled with the theft of every mundane and every significant joy from the life of another individual. It is easy to leave turkeys alone and let them have something to be thankful for as well.

Post

Previous Post

[The fish we kill to feed the fish we eat](#)

Comments

Jack Norris

Nov. 18, 2015, 4:07 p.m.

Thanks for your work clarifying this, Harish. I appreciate you taking the time to do the tedious work of bringing us accurate numbers!

Victoria Foote-Blackman

Nov. 18, 2015, 4:07 p.m.

Thank you so much, once again, for bringing your brilliant data crunch and analysis to this most troublesome subject of animal exploitation.

I will be sharing on my FB page.

Happy Thanksliving.

Rebecca Wolf

Nov. 18, 2015, 4:18 p.m.

Again and again,
Thank you for bringing numbers to movement. Your work is inspiring and invaluable!

John T. Maher

Nov. 18, 2015, 5:15 p.m.

Thanks Harish. The PRC maintains strategic stockpiles of frozen pork and chicken so they can manipulate price levels of these "commodities". Maybe frozen turkeys will follow. Are there futures contracts being arbitrated for Turkeys somewhere?

A lovely homily at the end para.

Emily Moran Barwick

Nov. 18, 2015, 5:26 p.m.

So well presented and researched, as always, Harish. Thank you so very much for your diligent work. It's so very important for these lives to be accounted for. Many thanks!

Emily Moran Barwick

Nov. 18, 2015, 5:28 p.m.

ps- Absolutely love your closing thoughts. I'll share this with my audience tomorrow.

Mikael

Nov. 18, 2015, 6:02 p.m.

Good stuff, yet again. Definitely sharing. Thanks, Harish :-)

Andrew Kirschner

Nov. 18, 2015, 8:20 p.m.

Outstanding and important work as always Harish.

Harish

Nov. 18, 2015, 9:20 p.m.

Thank you everyone for the very kind words!

John Maher: I do not know of any futures market in turkeys, but the cold storage data so far indicate that there may be 6-7% fewer turkeys eaten during Thanksgiving in 2015.

Jon Camp

Nov. 19, 2015, 1:02 p.m.

Interesting and useful as always, Harish. Thanks for continuing to keep your fellow advocates well-educated!

Berenice

Nov. 19, 2015, 2:52 p.m.

Thank you for all your great research! The numbers are always so overwhelming. At Truth or Drought, Lorelei Plotczyk does research on the California Drought, this meme for Thanksgiving <http://on.fb.me/1NePMqM>

karen@upc-online.org

Nov. 20, 2015, 1:54 p.m.

Thank you Harish for your important information and most importantly for reminding readers that we cannot truly give thanks for our own blessings while depriving turkeys and other fellow beings of theirs. Happy vegan Thanksgiving. Karen Davis, United Poultry Concerns <http://www.upc-online.org>

Maddie

Nov. 20, 2015, 2:13 p.m.

Harish,
Thank you for your incredible work. You provide us all with much needed data, which helps us all to be better as activists.

Mychael McNeeley

Nov. 20, 2015, 11:26 p.m.

Harish, I always eagerly await your posts. Thanks for all the work you put into this. 37 million is certainly better than 46 million, although nevertheless is truly staggering, especially when considering that we kill about as many turkeys for Thanksgiving alone as there are human citizens in my own state of California. It's truly a massacre of epic proportions, and all in the name of gratitude.

Additional comments via Facebook
